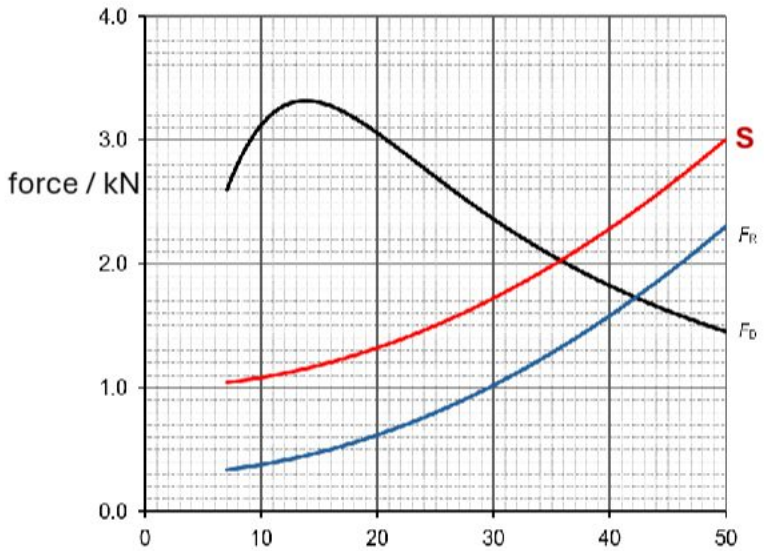
Paper 2

Question 1		
(a)(i)	12 to 14 m s^{-1} (The maximum acceleration occurs when the net force on the car = $F_D - F_R$ is largest.)	B1
(a)(ii)	Maximum resultant force on car, F_{net} = 3.30 kN – 0.40 kN (allow tolerance!) Maximum acceleration of the car = $\frac{(3.30-0.40)\times 10^3}{1200} = 2.42$ m s^{-2} (Allow ecf from 1(a) as long as F_{net} is > 0)	M1 A1
(a)(iii)	Rate of increase of the kinetic energy = $F_{net}\, v = (3.30 - 0.40) \times 10^3 \times 12$ = 3.48 x 10^4 W	C1 A1
(b)(i)	When going upslope, F_D needs to overcome both the F_R as well as the component of the weight parallel to the slope. Hence, the condition for zero net force occurs when F_D is larger and that occurs at a lower speed.	B1 B1
(b)(ii)	Correct shape – F_R shifted up by a constant value.  (The total force opposing the motion is now $F_R + mg \sin\theta$)	B1